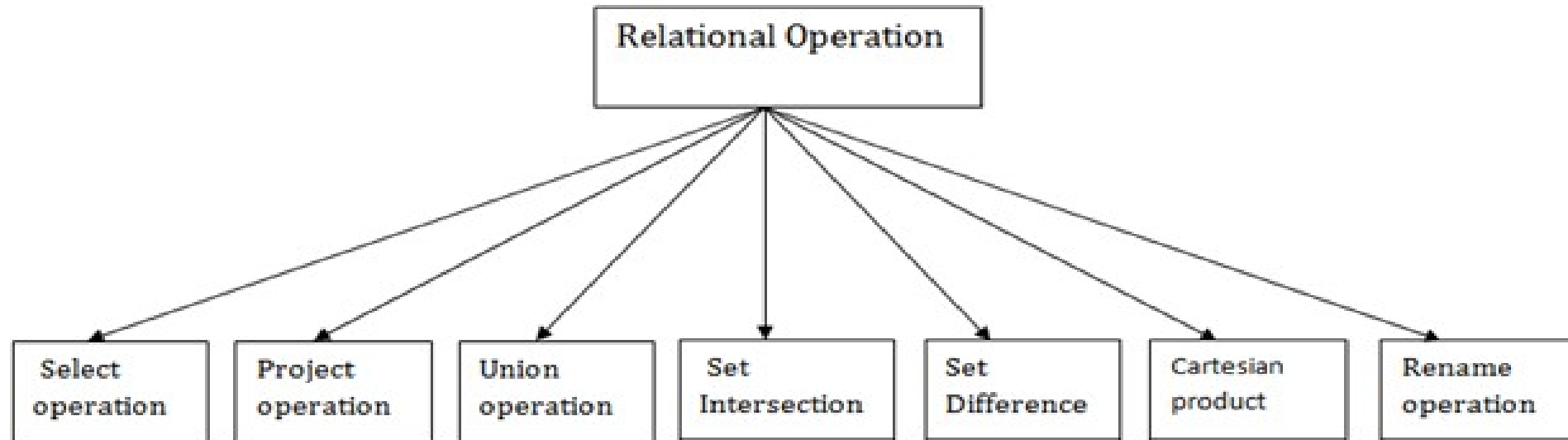


Relational Algebra

- Relational algebra is a procedural query language.
- It gives a step by step process to obtain the result of the query.
- It uses operators to perform queries.

Types of Relational operation



Select Operation

- Select operation selects tuples that satisfy a given predicate.
- It is denoted by sigma (σ).
- Notation: $\sigma p(r)$
- Where:
- σ is used for selection prediction
- r is used for relation
- p is used as a propositional logic formula which may use connectors like: AND OR and NOT.
- These relational can use as relational operators like $=, \neq, \geq, <, >, \leq$.

For example **Loan Relation**

BRANCH_NAME	LOAN_NO	AMOUNT
Downtown	L-17	1000
Redwood	L-23	2000
Perryride	L-15	1500
Downtown	L-14	1500
Mianus	L-13	500
Roundhill	L-11	900
Perryride	L-16	1300

Input:

σ BRANCH_NAME="perryride" (LOAN)

Output

BRANCH_NAME	LOAN_NO	AMOUNT
Perryride	L-15	1500
Perryride	L-16	1300

Project Operation

- This operation shows the list of those attributes that we wish to appear in the result.
- Rest of the attributes are eliminated from the table.
- It is denoted by Π .
- Notation: $\Pi A_1, A_2, A_n (r)$
- Where
- A_1, A_2, A_3 is used as an attribute name of relation r .

Example Customer Relation

Output

NAME	STREET	CITY
Jones	Main	Harrison
Smith	North	Rye
Hays	Main	Harrison
Curry	North	Rye
Johnson	Alma	Brooklyn
Brooks	Senator	Brooklyn

NAME	CITY
Jones	Harrison
Smith	Rye
Hays	Harrison
Curry	Rye
Johnson	Brooklyn
Brooks	Brooklyn

Input:

□ NAME, CITY (CUSTOMER)

Union Operation

- Suppose there are two tuples R and S. The union operation contains all the tuples that are either in R or S or both in R & S.
- It eliminates the duplicate tuples. It is denoted by $\cup$.
- Notation: $R \cup S$
- A union operation must hold the following condition:
- R and S must have the attribute of the same number.
- Duplicate tuples are eliminated automatically.

Example Depositor And Borrow Relation

Output

CUSTOMER_NAME	ACCOUNT_NO
Johnson	A-101
Smith	A-121
Mayes	A-321
Turner	A-176
Johnson	A-273
Jones	A-472
Lindsay	A-284

CUSTOMER_NAME	LOAN_NO
Jones	L-17
Smith	L-23
Hayes	L-15
Jackson	L-14
Curry	L-93
Smith	L-11
Williams	L-17

CUSTOMER_NAME
Johnson
Smith
Hayes
Turner
Jones
Lindsay
Jackson
Curry

Input:

$\sqcap$ CUSTOMER_NAME (BORROW) $\cup$ $\sqcap$ CUSTOMER_NAME (DEPOSITOR)

Set Intersection

- Suppose there are two tuples R and S. The set intersection operation contains all tuples that are in both R & S.
- It is denoted by intersection $\cap$.
- Notation: $R \cap S$
- Example: Using the above DEPOSITOR table and BORROW table

Example Depositor And Borrow Relation

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CUSTOMER_NAME	ACCOUNT_NO
Johnson	A-101
Smith	A-121
Mayes	A-321
Turner	A-176
Johnson	A-273
Jones	A-472
Lindsay	A-284

CUSTOMER_NAME	LOAN_NO
Jones	L-17
Smith	L-23
Hayes	L-15
Jackson	L-14
Curry	L-93
Smith	L-11
Williams	L-17

Output

CUSTOMER_NAME
Johnson
Smith

Input:

$\sqcap$ CUSTOMER_NAME (BORROW) $\cap$ $\sqcap$ CUSTOMER_NAME (DEPOSITOR)

Set Difference

- Suppose there are two tuples R and S. The set intersection operation contains all tuples that are in R but not in S.
- It is denoted by intersection minus (-).
- Notation: $R - S$
- **Example:** Using the above DEPOSITOR table and BORROW table

Example Depositor And Borrow Relation

Output

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CUSTOMER_NAME	ACCOUNT_NO
Johnson	A-101
Smith	A-121
Mayes	A-321
Turner	A-176
Johnson	A-273
Jones	A-472
Lindsay	A-284

CUSTOMER_NAME	LOAN_NO
Jones	L-17
Smith	L-23
Hayes	L-15
Jackson	L-14
Curry	L-93
Smith	L-11
Williams	L-17

CUSTOMER_NAME
Jackson
Hayes
Williams
Curry

Input:

Π CUSTOMER_NAME (BORROW) - Π CUSTOMER_NAME (DEPOSITOR)

Cartesian product

- The Cartesian product is used to combine each row in one table with each row in the other table. It is also known as a cross product.
- It is denoted by X.
- Notation: E X D

Example Employee and Department Output

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EMP_ID	EMP_NAME	EMP_DEPT
1	Smith	A
2	Harry	C
3	John	B

DEPT_NO	DEPT_NAME
A	Marketing
B	Sales
C	Legal

Input:

EMPLOYEE X DEPARTMENT

EMP_ID	EMP_NAME	EMP_DEPT	DEPT_NO	DEPT_NAME
1	Smith	A	A	Marketing
1	Smith	A	B	Sales
1	Smith	A	C	Legal
2	Harry	C	A	Marketing
2	Harry	C	B	Sales
2	Harry	C	C	Legal
3	John	B	A	Marketing
3	John	B	B	Sales
3	John	B	C	Legal

Rename Operation

- The rename operation is used to rename the output relation.
- It is denoted by **rho** (ρ).
- **Example:**
 - We can use the rename operator to rename STUDENT relation to STUDENT1.
- $\rho(\text{STUDENT1}, \text{STUDENT})$

Student

Roll No	Name	Adress

Student1

Roll No	Name	Adress